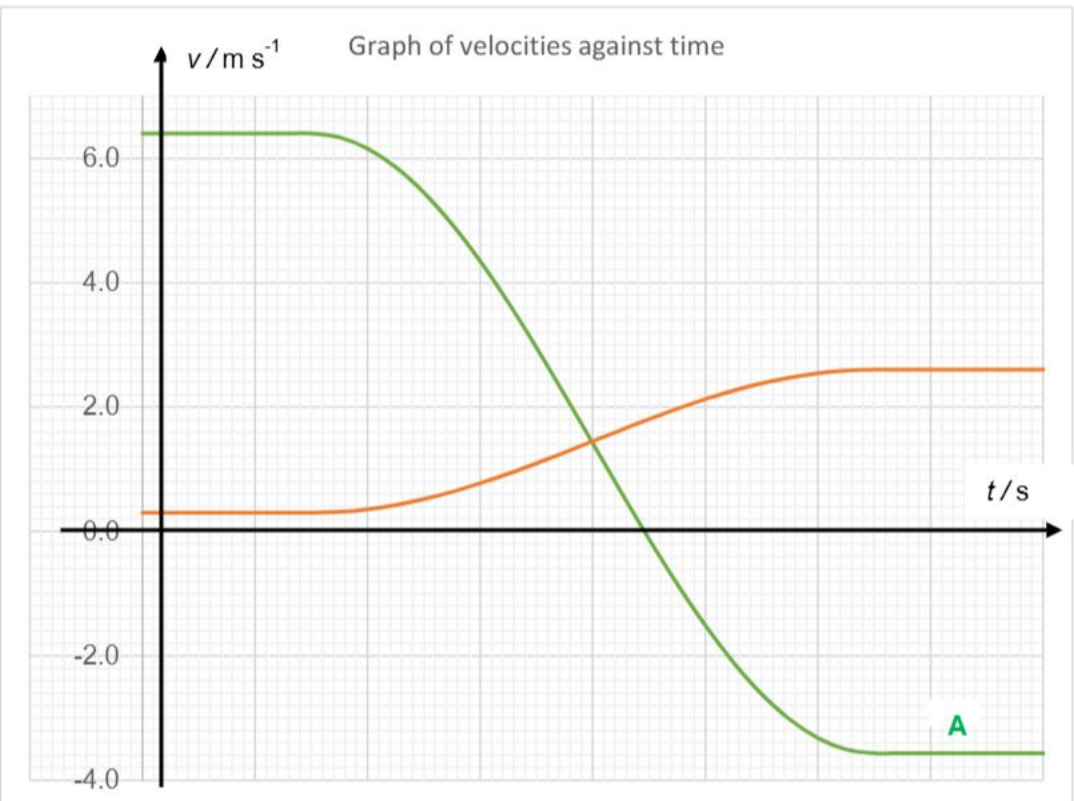
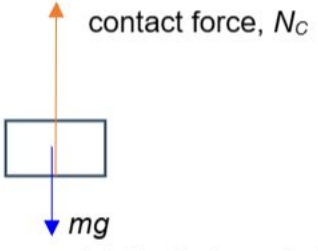
Question 9		
(a)	The **total linear momentum** of a system **remains unchanged** if **no net external force** acts on the system	**B1** **B1**
(b)(i)	Applying conservation of linear momentum: $m_A u_A + m_B u_B = m_A v_A + m_B v_B$ $(1.2)(6.40) + (5.2)(0.30) = 1.2v_A + 5.2v_B$ Since collision is elastic, relative speed of approach = relative speed of separation $u_A - u_B = v_B - v_A$ $v_A = v_B - 6.10$ Solving, $1.2v_A + 5.2(v_A + 6.10) = 9.24$ or $1.2(v_B - 6.10) + 5.2v_B = 9.24$ $v_B = 2.5875 = 2.6 \text{ m s}^{-1}$	**B1** **B1** **B1** **A0**
(b)(ii)	 From **(i)**: $v_A = 2.6 - 6.1 = -3.5 \text{ m s}^{-1}$ • Correct flat lines for where relevant • A smooth and downward sloping curve in between One mark is deducted for any mistake made	**A1** **A2**
(b)(iii)	Draw a tangent at the time on the curve of variation of velocity with time of trolley B. Determine its gradient and multiply it with the mass of trolley B to arrive at the force on trolley B.	**B1** **B1**

(b)(iv)	The total kinetic energy decreases upon collision and reaches a (non-zero) minimum value. After which, total kinetic energy increases and returns to its initial total kinetic energy before collision.	**B1** **B1**
(b)(v)	At maximum compression, the two trolleys have the same speed. Applying conservation of linear momentum: $m_A u_A + m_B u_B = (m_A + m_B)v_C$ $(1.2)(6.40) + (5.2)(0.30) = (1.2 + 5.2)v_C$ $v_C = 1.444 \text{ m s}^{-1}$ Applying conservation of energy Gain in E.P.E. = Loss in K.E. $= (0.5)(1.2)(6.40^2) + (0.5)(5.2)(0.30^2) - (0.5)(1.2 + 5.2)v_c^2$ $= 18.14$ $= 18.1 \text{ J}$ EPE in each spring = 9.1 J	**B1** **B1** **A1**
(c)(i)	v_Y is the vertical velocity of trollley before colliding with the ground, $v_y^2 = 2(9.81)(10.0)$ $v = 14.0 \text{ ms}^{-1}$ Average vertical force = $= \frac{m\Delta v}{\Delta t} = \frac{5.2(14.0 - 0)}{0.100} = 728 \text{ N}$ (upwards) Applying Newton's second law to the vertical direction: $N_C - mg = 728$ $N_C = 728 + 5.2(9.81) = 779 \text{ N}$  Note to marker: Award max 2 marks for students who ignore weight in their analysis.	**B1** **B1** **B1** **A1**
(c)(ii)	Impulse is the change in momentum, so impulse (in the vertical direction) is larger. (*Impulse in the horizontal direction depends on the angle of rebound. Students can discuss this but as long as their deduction makes sense.*)	**B1**